

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Code of Conduct:

I A. VAISHNAV (192524251) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. VAISHNAV (192524251)

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Understanding	20%	2	1	1	2
Method Selection	20%	1	2	1	2
Solution Process	25%	2	2	2	1
Accuracy of Calculation	20%	1	1	2	2
Interpretation of Results	15%	1	1	1	1

Total Marks 28 /40


Signature of the Faculty

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Q1. Subnetting : 192.168.20.0/24

Lab	Subnet	Network Address	Host Range	Broad Cast	Usable hosts
Lab A	126	192.168.20.0	192.168.20.1 - 192.168.20.62	192.168.20.63	62
Lab B	127	192.168.20.64	192.168.20.65 - 192.168.20.94	192.168.20.95	30
Lab C	128	192.168.20.96	192.168.20.97 - 192.168.20.110	192.168.20.111	14

Q2. Systems with 255.255.255.192 (126)

System 1:

- IP address : 192.168.1.130
- Network address : 192.168.1.128
- Host Range : 192.168.1.129 - 192.168.1.190
- Broadcast address : 192.168.1.191
- IP belongs to this subnet : yes
- Usable hosts : 62

System 2:

- IP address : 192.168.1.190
- Network address : 192.168.1.128
- Host Range : 192.168.1.129 - 192.168.1.190
- Broadcast address : 192.168.1.191
- IP belongs to this subnet : yes
- Usable hosts : 62

conclusion : Both systems are in the same subnet.

Q3. Subnetting : 172.16.0.0/16

Department	Subnet	Network add.	Host Range	Broadcast add.	usable hosts
HR	125	172.16.0.0	172.16.0.1 - 172.16.0.126	172.16.0.127	126
Sales	126	172.16.0.128	172.16.0.129 - 172.16.0.190	172.16.0.191	62
IT	127	172.16.0.192	172.16.0.193 - 172.16.0.222	172.16.0.223	30
admin	128	172.16.0.224	172.16.0.225 - 172.16.0.238	172.16.0.239	14

Remaining unused IP Range :-

172.16.0.240 - 172.16.255.255

Q4. 172.16.10.0/24 using 127

- Subnet Mask : 255.255.255.224
- No. of Subnets : 8
- usable hosts per subnet : 30

Subnet	Network	Broadcast
1	172.16.10.0	172.16.10.31
2	172.16.10.32	172.16.10.63
3	172.16.10.64	172.16.10.95
4	172.16.10.96	172.16.10.127
5	172.16.10.128	172.16.10.159
6	172.16.10.160	172.16.10.191
7	172.16.10.192	172.16.10.223

8

172.16.10.224

172.16.10.255

For IP: 172.16.10.78

- Network address : 172.16.10.64
- Host Range : 172.16.10.65 - 172.16.10.94
- Broadcast address : 172.16.10.95

Yes, 172.16.10.95 belongs to the same subnet but it is the broadcast address, not a usable host.

Q6.

IPv6Given address:-

2001:0db8:0000:0000:0000:ff00
:0042:8329.

compressed form:-

2001:db8:ff00:42:42:8329

Expanded form:-

2001:0db8:0000:0000:0000:
:ff00:0042:8329

Network prefix:-

2001:0db8:0000:0000::/64

No. of host address:-

$2^{64} = 18, 446, 744, 073, 709, 551, 616$